

A NASA Spacecraft Is About to Slingshot Past Mars at 12,000 MPH--and It's Taking Pictures!

From: ScienceDaily All Mined: 2026-05-11 3 min read

Have you ever spun a ball on a string and let it fly? That's kind of what NASA's Psyche spacecraft is about to do but with a whole planet! On May 15, Psyche will zip past Mars at a blistering 12,333 miles per hour, coming within 2,800 miles of the surface. It's not stopping for a visit. Instead, it will use Mars' gravity like a slingshot to bend its path and boost its speed, saving fuel on its long journey to a mysterious metal asteroid also named Psyche. The spacecraft launched in 2023 and cruises on a super-efficient solar-electric engine that puffs out xenon gas. But even that needs a little help. The Mars **flyby** is a **gravity assist** a clever trick that uses a planet's pull to change speed and direction without burning extra fuel. While it swings by, Psyche will be busy. Its camera will take thousands of pictures of Mars, starting with a thin crescent because it approaches from the night side. Scientists will use these images to practice for when it reaches the asteroid in 2029. They'll also test other instruments, like a **magnetometer** to feel Mars' magnetic field and a sensor to study **cosmic rays**. Other Mars spacecraft, like NASA's rovers and orbiters, will help track Psyche during the **flyby**. It's a team effort in space! The main goal is to get a gravity boost, but getting cool photos and valuable test data is, as one scientist put it, "the icing on the cake."

Word Treasure

gravity assist -- A clever space trick that uses a planet's gravity to speed up and change a spacecraft's path, like a giant slingshot.

flyby -- A close, quick pass by a planet or moon without stopping or landing.

xenon gas -- A heavy, invisible gas used in spacecraft engines to provide a gentle but steady push.

solar-electric engine -- A thruster that uses electricity from sunlight to push gas out the back, moving the spacecraft forward.

magnetometer -- An instrument that measures magnetic forces, like the invisible magnetic field around a planet.

cosmic rays -- High-energy particles that zoom through space and can tell scientists about faraway stars and galaxies.

Background you need

The asteroid Psyche is a 140-mile-wide world mostly made of iron and nickel, orbiting between Mars and Jupiter.

Gravity assists, also called slingshot maneuvers, were first used by the Mariner 10 spacecraft in 1974 to reach Mercury.

NASA's Perseverance rover on Mars sends data back to Earth and sometimes works with orbiters to track visiting spacecraft like Psyche.

Break it down

WHO: NASA's Psyche spacecraft.

WHAT: It will fly by Mars for a gravity assist to speed toward the asteroid Psyche.

WHERE: Past Mars, at a distance of about 2,800 miles from the surface.

WHY: To save fuel and test its cameras and instruments before reaching the asteroid in 2029.

Why it matters

Gravity assists let us explore deep space without carrying huge fuel tanks, helping us learn how planets are made--maybe even revealing secrets about Earth's own core.

Test what you remember

1. What is the name of the NASA spacecraft?
 - (A) Voyager
 - (B) Psyche
 - (C) Apollo
 - (D) Hubble
2. How fast will Psyche zip past Mars?
 - (A) 5,000 miles per hour
 - (B) 12,333 miles per hour
 - (C) 100,000 miles per hour
 - (D) 500 miles per hour
3. What does a gravity assist do?
 - (A) Uses a planet's gravity to change speed and direction without extra fuel
 - (B) Stops the spacecraft
 - (C) Makes the planet spin faster
 - (D) Creates a black hole
4. What will the camera do during the flyby?
 - (A) Turn off
 - (B) Take thousands of pictures of Mars
 - (C) Film a movie of Earth
 - (D) Snap a single photo each hour
5. Which instrument will measure Mars' magnetic field?
 - (A) Thermometer
 - (B) Magnetometer
 - (C) Telescope
 - (D) Microphone
6. What did one scientist call the photos and test data?
 - (A) The main dish
 - (B) The icing on the cake
 - (C) Dry toast
 - (D) A piece of pie

Answer key (try the quiz first!): 1: B 2: B 3: A 4: B 5: B 6: B

Questions to think about

1. **Innovation:** This zero-fuel slingshot trick makes long space trips possible and saves money for more science.
2. **Risk:** Any close approach to a planet has a tiny chance of a mistake, so mission teams must plan every detail very carefully.
3. **Scientific:** The test data from Mars will help scientists fine-tune the spacecraft's tools, making sure they work perfectly at the real target asteroid.

MY THOUGHTS (at least 20 words):